Grade 10, Term 2, Lesson 1, Criterion C1

Outcomes, Support and Probability Mass Functions

C1. Characterises discrete random variables in economic and financial contexts by identifying their support, probability mass function and possible outcomes.

All data are hypothetical teaching models. Keep exact expressions where useful and round final probabilities to four decimal places.

Highlight legend

Highlighted item	What it means
Outcome	A possible result of the experiment.
First variable	The first variable, its numerical values, and any units.
Second variable	The second variable, its numerical values, and any units.
Third variable	The third variable, its numerical values, and any units.
Probability	The chance of an outcome or event, expressed without units.
Unknown parameter	A quantity to determine from the information given.

Payment outcomes and their assigned values

A school shop receives one payment. The payment method is the *complete outcome*. A *random variable* assigns a number to each outcome.

Outcome ω	$P(\omega)$
Cash	0.1
Debit card	0.2
Credit card	0.3
Mobile wallet	0.4

Payment outcomes and their assigned values

A school shop receives one payment. The payment method is the *complete outcome*. A *random variable* assigns a number to each outcome.

Outcome ω	$P(\omega)$	$F(\omega)$ (USD)
Cash	0.1	0
Debit card	0.2	1
Credit card	0.3	1
Mobile wallet	0.4	2

F is the payment fee in USD.

Payment outcomes and their assigned values

A school shop receives one payment. The payment method is the *complete outcome*. A *random variable* assigns a number to each outcome.

Outcome ω	$P(\omega)$	$F(\omega)$ (USD)	$E(\omega)$
Cash	0.1	0	0
Debit card	0.2	1	1
Credit card	0.3	1	1
Mobile wallet	0.4	2	1

F is the payment fee in USD.
E is 1 for an electronic payment and 0 otherwise (unitless).

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C is 1 for a card payment and 0 otherwise (unitless).

Supports of the fee and indicators

The *support* is the set of values with positive probability.

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Cash	0.1	0	0	0
Debit card	0.2	1	1	1
Credit card	0.3	1	1	1
Mobile wallet	0.4	2	1	0

F is the payment fee in USD.
E is 1 for an electronic payment and 0 otherwise (unitless).
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Predict. Which distinct values appear in each variable's column?

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Mobile wallet	0.4	2	1	0

F is the payment fee in USD.
E is 1 for an electronic payment and 0 otherwise (unitless).
C is 1 for a card payment and 0 otherwise (unitless).

Predict. Which distinct values appear in each variable's column?

F : $\{0, 1, 2\}$ (USD) E : $\{0, 1\}$ (unitless) C : $\{0, 1\}$ (unitless)

Probability of a 1 USD fee

A *probability mass function* (PMF) gives each value's probability.

Outcome ω	$P(\omega)$	$F(\omega)$ (USD)	$E(\omega)$	$C(\omega)$
Cash	0.1	0	0	0
Debit card	0.2	1	1	1
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Mobile wallet	0.4	2	1	0

F is the payment fee in USD.

Predict. Which outcomes give $F = 1$?

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F is the payment fee in USD.

Predict. Which outcomes give $F = 1$?

Debit card and Credit card are mutually exclusive, so add their probabilities.

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Debit card	0.2	1	1	1
Credit card	0.3	1	1	1
Mobile wallet	0.4	2	1	0

F is the payment fee in USD.

Predict. Which outcomes give $F = 1$?

Debit card and Credit card are mutually exclusive, so add their probabilities.

$$P(F = 1) = 0.2 + 0.3 = 0.5.$$

Probability of an electronic payment

Outcome ω	$P(\omega)$	$F(\omega)$ (USD)	$E(\omega)$	$C(\omega)$
Cash	0.1	0	0	0
Debit card	0.2	1	1	1
Credit card	0.3	1	1	1
Mobile wallet	0.4	2	1	0

E is 1 for an electronic payment and 0 otherwise (unitless).

Predict. Which outcomes give $E = 1$?

Probability of an electronic payment

Outcome ω	$P(\omega)$	$F(\omega)$ (USD)	$E(\omega)$	$C(\omega)$
Cash	0.1	0	0	0
Debit card	0.2	1	1	1
Credit card	0.3	1	1	1
Mobile wallet	0.4	2	1	0

E is 1 for an electronic payment and 0 otherwise (unitless).

Predict. Which outcomes give $E = 1$?
Debit card, Credit card and Mobile wallet.

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Outcome ω	$P(\omega)$	$F(\omega)$ (USD)	$E(\omega)$	$C(\omega)$
Cash	0.1	0	0	0
Debit card	0.2	1	1	1
Credit card	0.3	1	1	1
Mobile wallet	0.4	2	1	0

E is 1 for an electronic payment and 0 otherwise (unitless).

Predict. Which outcomes give $E = 1$?
Debit card, Credit card and Mobile wallet.

$$P(E = 1) = 0.2 + 0.3 + 0.4 = 0.9.$$

Probability of a card payment

Outcome ω	$P(\omega)$	$F(\omega)$ (USD)	$E(\omega)$	$C(\omega)$
Cash	0.1	0	0	0
Debit card	0.2	1	1	1
Credit card	0.3	1	1	1
Mobile wallet	0.4	2	1	0

C is 1 for a card payment and 0 otherwise (unitless).

Predict. Which outcomes give $C = 1$?

Probability of a card payment

Outcome ω	$P(\omega)$	$F(\omega)$ (USD)	$E(\omega)$	$C(\omega)$
Cash	0.1	0	0	0
Debit card	0.2	1	1	1
Credit card	0.3	1	1	1
Mobile wallet	0.4	2	1	0

C is 1 for a card payment and 0 otherwise (unitless).

Predict. Which outcomes give $C = 1$?
Debit card and Credit card.

Probability of a card payment

Outcome ω	$P(\omega)$	$F(\omega)$ (USD)	$E(\omega)$	$C(\omega)$
Cash	0.1	0	0	0
Debit card	0.2	1	1	1
Credit card	0.3	1	1	1
Mobile wallet	0.4	2	1	0

C is 1 for a card payment and 0 otherwise (unitless).

Predict. Which outcomes give $C = 1$?
Debit card and Credit card.

$$P(C = 1) = 0.2 + 0.3 = 0.5.$$

The fee PMF

Outcome ω	$P(\omega)$	$F(\omega)$ (USD)	$E(\omega)$	$C(\omega)$
Cash	0.1	0	0	0
Debit card	0.2	1	1	1
Credit card	0.3	1	1	1
Mobile wallet	0.4	2	1	0

Fee value f (USD)	0	1	2
$P(F = f)$	0.1	0.5	0.4

Worksheet reading. See the worked grouping problem for the full fee calculation.

The electronic-payment and card-payment PMFs

E is 1 for an electronic payment and 0 otherwise (unitless).
C is 1 for a card payment and 0 otherwise (unitless).

Electronic-payment value e	0	1
$P(E = e)$	0.1	0.9
Card-payment value c	0	1
$P(C = c)$	0.5	0.5

E and C share a support but have different masses. All probabilities are unitless.

Three independent visitor decisions

A school enterprise offers a 5 USD notebook to each of three visitors.

Decision	Symbol	Probability
Buy	B	0.4
Pass	N	0.6

Assume the visitors decide *independently*. One visitor's decision leaves the other visitors' probabilities the same.

Eight ordered outcomes

Choose `B` or `N` for each visitor. Read the three positions in visitor order.

Predict. List the complete outcomes with zero, one, two and three buyers.

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Eight ordered outcomes

Choose B or N for each visitor. Read the three positions in visitor order.

Predict. List the complete outcomes with zero, one, two and three buyers.

$$\Omega = \{\mathrm{NNN}, \mathrm{BNN}, \mathrm{NBN}, \mathrm{NNB}, \mathrm{BBN}, \mathrm{BNB}, \mathrm{NBB}, \mathrm{BBB}\}.$$

Probability that only the first visitor buys

$$\Omega = \{\mathrm{NNN}, \mathrm{BNN}, \mathrm{NBN}, \mathrm{NNB}, \mathrm{BBN}, \mathrm{BNB}, \mathrm{NBB}, \mathrm{BBB}\}.$$

In BNN, only the first visitor buys. Independence lets us multiply the three decision probabilities in visitor order.

Predict. Which factor belongs to each position in BNN?

$$P(\,\mathrm{BNN}\,)$$

Probability that only the first visitor buys

$$\Omega = \{\mathrm{NNN}, \mathrm{BNN}, \mathrm{NBN}, \mathrm{NNB}, \mathrm{BBN}, \mathrm{BNB}, \mathrm{NBB}, \mathrm{BBB}\}.$$

In BNN, only the first visitor buys. Independence lets us multiply the three decision probabilities in visitor order.

Predict. Which factor belongs to each position in BNN?

$$P(\mathrm{BNN}) = 0.4$$

Probability that only the first visitor buys

$$\Omega = \{\mathrm{NNN}, \mathrm{BNN}, \mathrm{NBN}, \mathrm{NNB}, \mathrm{BBN}, \mathrm{BNB}, \mathrm{NBB}, \mathrm{BBB}\}.$$

In BNN, only the first visitor buys. Independence lets us multiply the three decision probabilities in visitor order.

Predict. Which factor belongs to each position in BNN?

$$P(\mathrm{BNN}) = 0.4 \times 0.6$$

Probability that only the first visitor buys

$$\Omega = \{\mathrm{NNN}, \mathrm{BNN}, \mathrm{NBN}, \mathrm{NNB}, \mathrm{BBN}, \mathrm{BNB}, \mathrm{NBB}, \mathrm{BBB}\}.$$

In BNN, only the first visitor buys. Independence lets us multiply the three decision probabilities in visitor order.

Predict. Which factor belongs to each position in BNN?

$$P(\mathrm{BNN}) = 0.4 \times 0.6 \times 0.6$$

Probability that only the first visitor buys

$$\Omega = \{\mathrm{NNN}, \mathrm{BNN}, \mathrm{NBN}, \mathrm{NNB}, \mathrm{BBN}, \mathrm{BNB}, \mathrm{NBB}, \mathrm{BBB}\}.$$

In BNN, only the first visitor buys. Independence lets us multiply the three decision probabilities in visitor order.

Predict. Which factor belongs to each position in BNN?

$$P(\mathrm{BNN}) = 0.4 \times 0.6 \times 0.6 = 0.144.$$

Outcomes with the same probability product

$$\Omega = \{\text{NNN}, \text{BNN}, \text{NBN}, \text{NNB}, \text{BBN}, \text{BNB}, \text{NBB}, \text{BBB}\}.$$

Predict. Which outcomes have the same factors, possibly in a different order?

Complete outcomes	Probability of each outcome

Outcomes with the same probability product

$$\Omega = \{\text{NNN}, \text{BNN}, \text{NBN}, \text{NNB}, \text{BBN}, \text{BNB}, \text{NBB}, \text{BBB}\}.$$

Predict. Which outcomes have the same factors, possibly in a different order?

Complete outcomes	Probability of each outcome
NNN	
BNN, NBN, NNB	
BBN, BNB, NBB	
BBB	

Outcomes with the same probability product

$$\Omega = \{\text{NNN}, \text{BNN}, \text{NBN}, \text{NNB}, \text{BBN}, \text{BNB}, \text{NBB}, \text{BBB}\}.$$

Predict. Which outcomes have the same factors, possibly in a different order?

Complete outcomes	Probability of each outcome
NNN	$0.6^3 = 0.216$
BNN, NBN, NNB	$0.4 \times 0.6 \times 0.6 = 0.144$
BBN, BNB, NBB	$0.4^2 \times 0.6 = 0.096$
BBB	$0.4^3 = 0.064$

Buyer count, revenue and first visitor

$$\Omega = \{\mathrm{NNN}, \mathrm{BNN}, \mathrm{NBN}, \mathrm{NNB}, \mathrm{BBN}, \mathrm{BNB}, \mathrm{NBB}, \mathrm{BBB}\}.$$

Variable	Finite support	Units
X counts the buyers among the three visitors.	$\{0, 1, 2, 3\}$	buyers
$Y = 5X$ is total notebook revenue.	$\{0, 5, 10, 15\}$	USD
Z is 1 if the first visitor buys and 0 if the first visitor passes.	$\{0, 1\}$	Unitless

Comparing BNN and NBN

In visitor order, B means buy and N means pass.
X counts the buyers among the three visitors.
$Y = 5X$ is total notebook revenue in USD.
Z is 1 if the first visitor buys and 0 if the first visitor passes (unitless).

Predict. Which values agree for these two outcomes?

Outcome	X (buyers)	Y (USD)	Z (unitless)
BNN			
NBN			

Comparing BNN and NBN

In visitor order, B means buy and N means pass.
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$Y = 5X$ is total notebook revenue in USD.
Z is 1 if the first visitor buys and 0 if the first visitor passes (unitless).

Predict. Which values agree for these two outcomes?

Outcome	X (buyers)	Y (USD)	Z (unitless)
BNN	1	5	
NBN	1	5	

Both outcomes have one buyer, so the count and revenue agree.

Comparing BNN and NBN

In visitor order, B means buy and N means pass.
X counts the buyers among the three visitors.
$Y = 5X$ is total notebook revenue in USD.
Z is 1 if the first visitor buys and 0 if the first visitor passes (unitless).

Predict. Which values agree for these two outcomes?

Outcome	X (buyers)	Y (USD)	Z (unitless)
BNN	1	5	1
NBN	1	5	0

Both outcomes have one buyer, so the count and revenue agree.

The first visitor buys in BNN and passes in NBN, so Z differs.

Probability of exactly one buyer

X counts the buyers among the three visitors.
In visitor order, B means buy (0.4) and N means pass (0.6).

$$\Omega = \{\mathrm{NNN}, \mathrm{BNN}, \mathrm{NBN}, \mathrm{NNB}, \mathrm{BBN}, \mathrm{BNB}, \mathrm{NBB}, \mathrm{BBB}\}.$$

Predict. Which outcomes give $X = 1$?

Probability of exactly one buyer

X counts the buyers among the three visitors.
In visitor order, B means buy (0.4) and N means pass (0.6).

$$\Omega = \{\mathrm{NNN}, \mathrm{BNN}, \mathrm{NBN}, \mathrm{NNB}, \mathrm{BBN}, \mathrm{BNB}, \mathrm{NBB}, \mathrm{BBB}\}.$$

Predict. Which outcomes give $X = 1$?
Add the probabilities of BNN, NBN and NNB. These complete outcomes are mutually exclusive.

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Predict. Which outcomes give $X = 1$?
Add the probabilities of BNN, NBN and NNB. These complete outcomes are mutually exclusive.

$$P(X = 1) = P(\mathrm{BNN}) + P(\mathrm{NBN}) + P(\mathrm{NNB})$$

Probability of exactly one buyer

X counts the buyers among the three visitors.
In visitor order, B means buy (0.4) and N means pass (0.6).

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Predict. Which outcomes give $X = 1$?
Add the probabilities of BNN, NBN and NNB. These complete outcomes are mutually exclusive.

$$\begin{aligned} P(X = 1) &= P(\mathrm{BNN}) + P(\mathrm{NBN}) + P(\mathrm{NNB}) \\ &= 0.144 + 0.144 + 0.144 = 0.432. \end{aligned}$$

Worksheet reading. The worked school-sale problem completes all three PMFs. The notation reference shows how to write and check the sums.

Probability of 10 USD in revenue

X counts the buyers among the three visitors.
$Y = 5X$ is total notebook revenue in USD.
In visitor order, B means buy (0.4) and N means pass (0.6).

$$\Omega = \{\mathrm{NNN}, \mathrm{BNN}, \mathrm{NBN}, \mathrm{NNB}, \mathrm{BBN}, \mathrm{BNB}, \mathrm{NBB}, \mathrm{BBB}\}.$$

Predict. Which outcomes give $Y = 10$?

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$$\Omega = \{\mathrm{NNN}, \mathrm{BNN}, \mathrm{NBN}, \mathrm{NNB}, \mathrm{BBN}, \mathrm{BNB}, \mathrm{NBB}, \mathrm{BBB}\}.$$

Predict. Which outcomes give $Y = 10$?
At 5 USD per notebook, 10 USD means $10 \div 5 = 2$ buyers. The matches are BBN, BNB and NBB.

Probability of 10 USD in revenue

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Predict. Which outcomes give $Y = 10$?
At 5 USD per notebook, 10 USD means $10 \div 5 = 2$ buyers. The matches are BBN, BNB and NBB.

$$P(Y = 10) = P(\mathrm{BBN}) + P(\mathrm{BNB}) + P(\mathrm{NBB})$$

Independence lets us multiply the decision probabilities within an outcome. The matching outcomes are mutually exclusive, so we add their probabilities.

Probability of 10 USD in revenue

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Predict. Which outcomes give $Y = 10$?
At 5 USD per notebook, 10 USD means $10 \div 5 = 2$ buyers. The matches are BBN, BNB and NBB.

$$\begin{aligned} P(Y = 10) &= P(\mathrm{BBN}) + P(\mathrm{BNB}) + P(\mathrm{NBB}) \\ &= 0.096 + 0.096 + 0.096 \end{aligned}$$

Independence lets us multiply the decision probabilities within an outcome. The matching outcomes are mutually exclusive, so we add their probabilities.

Probability of 10 USD in revenue

X counts the buyers among the three visitors.
$Y = 5X$ is total notebook revenue in USD.
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Predict. Which outcomes give $Y = 10$?
At 5 USD per notebook, 10 USD means $10 \div 5 = 2$ buyers. The matches are BBN, BNB and NBB.

$$\begin{aligned} P(Y = 10) &= P(\mathrm{BBN}) + P(\mathrm{BNB}) + P(\mathrm{NBB}) \\ &= 0.096 + 0.096 + 0.096 = 0.2880. \end{aligned}$$

Independence lets us multiply the decision probabilities within an outcome. The matching outcomes are mutually exclusive, so we add their probabilities.

Probability for the first visitor

Z is 1 if the first visitor buys and 0 if the first visitor passes (unitless).
In visitor order, B means buy (0.4) and N means pass (0.6).

$$\Omega = \{\mathrm{NNN}, \mathrm{BNN}, \mathrm{NBN}, \mathrm{NNB}, \mathrm{BBN}, \mathrm{BNB}, \mathrm{NBB}, \mathrm{BBB}\}.$$

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Predict. Which outcomes give $Z = 1$?
Only the first position must be B. The other two visitors may buy or pass. The matches are BNN, BBN, BNB and BBB.

Probability for the first visitor

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In visitor order, B means buy (0.4) and N means pass (0.6).

$$\Omega = \{\mathrm{NNN}, \mathrm{BNN}, \mathrm{NBN}, \mathrm{NNB}, \mathrm{BBN}, \mathrm{BNB}, \mathrm{NBB}, \mathrm{BBB}\}.$$

Predict. Which outcomes give $Z = 1$?
Only the first position must be B. The other two visitors may buy or pass. The matches are BNN, BBN, BNB and BBB.

$$P(Z = 1) = P(\mathrm{BNN}) + P(\mathrm{BBN}) + P(\mathrm{BNB}) + P(\mathrm{BBB})$$

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Predict. Which outcomes give $Z = 1$?
Only the first position must be B. The other two visitors may buy or pass. The matches are BNN, BBN, BNB and BBB.

$$\begin{aligned} P(Z = 1) &= P(\mathrm{BNN}) + P(\mathrm{BBN}) + P(\mathrm{BNB}) + P(\mathrm{BBB}) \\ &= 0.144 + 0.096 + 0.096 + 0.064 \end{aligned}$$

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Probability for the first visitor

Z is 1 if the first visitor buys and 0 if the first visitor passes (unitless).
In visitor order, B means buy (0.4) and N means pass (0.6).

$$\Omega = \{\mathrm{NNN}, \mathrm{BNN}, \mathrm{NBN}, \mathrm{NNB}, \mathrm{BBN}, \mathrm{BNB}, \mathrm{NBB}, \mathrm{BBB}\}.$$

Predict. Which outcomes give $Z = 1$?
Only the first position must be B. The other two visitors may buy or pass. The matches are BNN, BBN, BNB and BBB.

$$\begin{aligned} P(Z = 1) &= P(\mathrm{BNN}) + P(\mathrm{BBN}) + P(\mathrm{BNB}) + P(\mathrm{BBB}) \\ &= 0.144 + 0.096 + 0.096 + 0.064 = 0.4000. \end{aligned}$$

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In visitor order, B means buy (0.4) and N means pass (0.6).

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Predict. Which outcomes give $Z = 1$?
Only the first position must be B. The other two visitors may buy or pass. The matches are BNN, BBN, BNB and BBB.

$$\begin{aligned} P(Z = 1) &= P(\mathrm{BNN}) + P(\mathrm{BBN}) + P(\mathrm{BNB}) + P(\mathrm{BBB}) \\ &= 0.144 + 0.096 + 0.096 + 0.064 = 0.4000. \end{aligned}$$

Independence lets us multiply the decision probabilities within an outcome. The matching outcomes are mutually exclusive, so we add their probabilities.

This equals the original 0.4 probability that the first visitor buys.

Discrete supports and purchase time

Braces list the values in a support.

Variable	Finite support	Units
X counts the buyers among the three visitors.	$\{0, 1, 2, 3\}$	buyers
$Y = 5X$ is total notebook revenue.	$\{0, 5, 10, 15\}$	USD
Z is 1 if the first visitor buys and 0 if the first visitor passes.	$\{0, 1\}$	Unitless

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A **countably infinite** support has infinitely many values that can be listed one by one, as in $\{1, 2, 3, \ldots\}$.

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The **exact time to complete a purchase** in seconds is continuous. It can take values throughout an interval.

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A **countably infinite** support has infinitely many values that can be listed one by one, as in $\{1, 2, 3, \ldots\}$.

The **exact time to complete a purchase** in seconds is continuous. It can take values throughout an interval.

Purchase time recorded in whole seconds takes discrete values.